

Article

Education, Science and Technology, and Talent Integrated Development: Evidence from China

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Abstract: Education, science and technology, and talent are significant factors driving economic growth. Coordinating the integrated development of education, science and technology, and talent is not only an important way to achieve the development of education, technology, and talent themselves, but it is also a crucial path to sustainable development. The research objectives of this paper are to explore the coupling coordination relationship among the three subsystems of education, science and technology, and talent and to investigate their spatiotemporal pattern evolution characteristics. This paper, based on panel data from 30 provinces in China from 2001 to 2022, constructs an evaluation index system for “education–science and technology–talent” (EST) development from three dimensions: education, science and technology, and talent. It comprehensively employs methods such as spatiotemporal pattern evolution, LISA time path, and spatial transitions to evaluate the level of China’s EST. The results indicate the following: (1) During the research period, the level of China’s EST shows a significant upward trend, with the educational subsystem being the most prominent. (2) The level of China’s EST development has a significant spatial differentiation characteristic, generally showing a decreasing trend from the eastern to the central and western regions. (3) In terms of the stability of the spatiotemporal pattern, the spatial pattern of China’s EST development is relatively stable, and there is no significant spatial shift during the research period. This study provides a new method for the evaluation of complex systems and also offers a reference for the economic growth of many developing countries.

Keywords: education; science and technology; talent; spatiotemporal characteristic; China



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1. Introduction

In development economics, education, technology, and talent have always been regarded as an essential driving force of economic growth [1]. Education, as an important means of achieving social mobility, has been widely valued by all social strata [2]. Given the varying family situations of each student, the level of education naturally possesses an attribute of inequality. Technological innovation is a crucial way to achieve increasing returns to scale, giving rise to a multitude of new industries, models, and scenarios, thereby driving economic growth. The transformation and application of innovation achievements are key to integrating scientific research with economic output, and they represent the “last mile” of achieving innovation-driven development [3]. Technological innovation itself does not directly generate economic benefits; it must rely on adequate industrial investment to create suitable application scenarios for the transformation of scientific and technological achievements [4]. The application scenario-driven technological innovation process is the most practical and demand-fitting innovation process, where science and technology are further enhanced. Talent is an important and fundamental resource, and the development of education and technology is supported and driven by talent.

The existing literature explored the economic consequences of education, technology, and talent. Based on the existing literature, some studies further investigated the interrelationships between them. (1) In terms of the relationship between technology and talent, technological development is an important manifestation of the continuous improvement in education levels and the aggregation of talent. Researchers are a significant driving force in advancing scientific and technological progress, and the development of science and technology requires researchers to continuously exercise their initiative and creativity. Providing funding for basic research is an important way to achieve technological development and talent cultivation, especially having a positive effect on the research output and long-term development of young scientific and technological personnel [5]. In corporate innovation activities, not only is there a need for a large number of professional researchers engaged in research and development activities, but there is also a need for a large number of professional managers. A significant number of young management talents have a marked positive effect on corporate radical innovation [6]. (2) In terms of the relationship between education and talent, labor market supply policies alone are not sufficient to cope with market changes; it is necessary to use a broader range of education and training to continuously improve the level of human capital. It is essential to further increase investment in scientific and technological innovation; plan tax policies with a focus on workers and social challenges, to further activate the scientific and technological innovation system; and enhance the overall efficiency of the operation of the scientific and technological innovation system [7]. As an important sign of a new generation of science and technology, artificial intelligence technology has developed rapidly in recent years, and the demand for high-level talents has been increasing. Various artificial intelligence technologies, while replacing low-skilled labor, also set higher demands for high-skilled workers. Researchers point out that cooperation between artificial intelligence and employees with high task experience and high sensitivity to artificial intelligence will have better production efficiency [8]. Due to the inconsistency in education levels, the redistribution gap of college graduates worldwide has further widened, exacerbating the extent of talent loss in developing countries [9]. (3) In terms of the relationship between education and technology, there are significant differences in the university-educated labor force between different countries, and education has a significant impact on the quality of human capital. Some studies point out that an important standard of regional education level—the quality of college graduates—is crucial for a country's economic development [9,10]. For every one-standard-deviation increase, the proportion of graduates becoming inventors almost doubles, and the number of Nobel laureates and entrepreneurs increases by approximately 0.1 and 1%, respectively.

However, the above research on EST has some shortcomings. Firstly, existing studies have individually examined the economic consequences brought about by the improvement in education, technology, and talent levels and discussed the relationships between education, technology, and talent. However, as the interactions between the subsystems of education, technology, and talent increase, the interdependent and mutually restrictive relationships among the three have become more evident, urgently necessitating research on the relationships among them. Secondly, in terms of data construction, existing studies have widely adopted relevant statistical data from the National Bureau of Statistics of China and various provinces, with relatively singular data sources and not achieving the comprehensive use of multiple data sources in articles. This leads to an insufficient reflection of the evaluation results on reality, making it difficult to achieve cross-verification between different research outcomes. Thirdly, research on the spatial and temporal differences in complex systems often remains at the stage of using spatial and temporal thinking to analyze the geographical imbalance of their development levels. However, such analysis often employs exploratory spatiotemporal data analysis methods, resulting in analysis that is limited to the spatial dimension and does not unify time and space within a single analytical framework.

Although existing research has made certain progress, the coupling coordination relationship among the three subsystems of education, science and technology, and talent has

not been fully and clearly explained. Therefore, this paper focuses on the coupling coordination level of the three subsystems—education, science and technology, and talent—and conducts research from the perspective of economic geography to explore the coupling coordination mechanisms and spatial pattern characteristics of these three elements. To remedy the deficiencies mentioned above, (1) based on existing research on education, technology, talent, and their relationships, this paper focuses on the coordinated characteristics of high-quality economic development in China, starting from the perspective of systems theory, to theoretically analyze the interrelationships among education, technology, and talent, and on this basis, it constructs an evaluation index system for the integrated development level of EST. (2) In terms of index system construction and development level measurement, this paper constructs a database for measuring the integrated level of EST, comprehensively employing various data analysis methods and using multiple sources of data such as statistical data, remote sensing data, and text analysis data to measure the integrated development level of EST. (3) In spatial analysis, this paper constructs an asymmetric spatial vector weight matrix that can reflect the economic–geographical two-dimensional distance. On this basis, an exploratory spatiotemporal data analysis model that incorporates the temporal dimension is built based on the exploratory spatial data analysis model, achieving the coupling of time and space on the weight matrix.

2. Theoretical Framework and Analysis of Coupling Mechanism

EST, as an important support for the high-quality economic development of China, plays an irreplaceable role and function. From the perspective of systems theory, the input and output relationships formed by the independent operation of the education, technology, and talent systems not only provide the necessary material basis and conditions for the normal operation of the system but also create the necessary prerequisites for the three to jointly constitute a complex megasystem in the complex network of interrelationships [11].

Within the EST system, the educational subsystem holds a foundational position. Throughout the development of human society, the role of education has never been negligible. Firstly, education has a unique role in teaching and nurturing individuals. Besides personal practice, education is an important channel for acquiring knowledge. Through education, individuals can gain the knowledge they need. Teachers, by teaching various types of courses, facilitate the interaction between different bodies of knowledge and the formation of knowledge networks. This process not only allows for the acquisition of necessary knowledge but also enhances personal qualities, thereby promoting the advancement of human civilization [12]. Secondly, education serves as a medium for the diffusion of knowledge. When a piece of knowledge is confined to an individual and has not yet been recognized by the scientific community or is not well known to the public, the mediating role of education becomes extremely important. The implementation of education is essentially the process of disseminating consensus-based knowledge. This process enables the transfer and diffusion of knowledge among different groups and across generations, thereby providing significant assistance in the popularization of a particular piece of knowledge [13].

The science and technology subsystem plays a dynamic role within EST. Science and technology are the most important driving forces of economic and social development. Firstly, the development of the technological subsystem requires a large number of high-quality talents [14]. The advancement of technological innovation is a process of going from 0 to 1, where there is no ready-made experience to draw upon; thus, it must rely on high-quality talents to use their innovative thinking for cutting-edge exploration [15]. The application and transformation of innovative results need to be closely integrated with practice, achieving a combination of theoretical innovation and applied innovation, as well as technological innovation and industrial innovation. A large number of industrial engineers and technical workers promote the implementation of innovative results during this stage [16]. Secondly, the development of the technological subsystem requires education as an important support. In the process of educational development, the empowerment of

technology is crucial. Science and technology permeate all fields and the entire process of economic and social development, with the Internet closely integrated with social production activities. This has cultivated a group of compound talents who understand both the application of scientific and technological knowledge and the development of industrial technology, significantly promoting the modernization of education and the renewal of talent training models [17].

The talent subsystem occupies a central position within EST. The progress of society is closely related to people, which also determines that people are the core of EST. High-skilled laborers who have undergone rigorous thinking training are an important part of the talent subsystem and play a significant role in enhancing the overall level of the talent subsystem [18,19]. They become the force urgently needed in education. Whether it is education or science and technology, the full participation of people and the full play of their subjective initiative are required. Education is an activity with people as the main body. Teachers, students, administrators, and others are all participants in the educational process, each playing a unique role in the dissemination of knowledge. In the field of technological innovation, high-level technological innovation often achieves significant breakthroughs from 0 to 1, which requires relying on education to cultivate talents with rich knowledge and innovation and relying on research teams of scientists and researchers with extremely high quality to tackle key issues.

In order to better illustrate the coupling mechanism among education, science and technology, and talent, our paper further analyzes the relationships between points, lines, planes, and solids in geometry (Figure 1). In the three-dimensional coordinate system of Figure 1, the cube can represent the level of coordinated development of EST. Plane_{AOB}, Plane_{AOC}, and Plane_{BOC} represent the coupling and coordination states between any two subsystems of education, science and technology, and talent, respectively. For ease of analysis and discussion, this paper assumes that Plane_{AOB} is the coupling and coordination state of the education and science and technology subsystems, and Line_{OC} represents the development level of the talent subsystem. Arrow_a, Arrow_b, and Arrow_c represent the coupling and coordination development levels of the education and science and technology subsystems at different times, respectively, and there are countless lines within Plane_{AOB} representing the coupling and coordination development states of the education and science and technology subsystems at several times. The farther the endpoint of the line within the plane is from Point_O, the higher the coupling and coordination development level of the education and science and technology subsystems. Similarly, it can be concluded that Plane_{AOC} and Plane_{BOC} represent the coupling and coordination development states of any two subsystems. As the development level of any subsystem continues to improve, the development level of EST can be represented by any line within the spatial range, and the farther the endpoint of the line within the space is from Point_O, the higher the coupling and coordination development level of EST. Spatial rays w_1 , w_2 , and w_3 represent the coupling and coordination development levels of EST, where $w_1 < w_2 < w_3$. The interdependent and mutually influential “game” relationship among the education, science and technology, and talent subsystems together form the basis for the system’s operation and development, and the interaction coupling between any two systems determines the direction of the system’s progress. When the ray emanating from Point_O is in the optimal position, it means that the coupling and coordination among the education, science and technology, and talent subsystems have reached the best state.

From a spatial perspective, due to significant differences in resource endowments, social environments, and knowledge accumulation across various regions, the integrated development of the education, science and technology, and talent systems exhibits varying levels of development at different spatial scales. They possess distinct spatial differentiation characteristics. Based on this, once the dimension of temporal evolution is incorporated, the coupling state of EST and the local and adjacent relationships will undergo corresponding changes at different points in time. The types of spatial correlations and the transition states of the coupling coordination degree also change accordingly, together forming the

spatial structure, distribution patterns, and temporal sequence evolution characteristics of the integrated development of EST under spatiotemporal transition states.

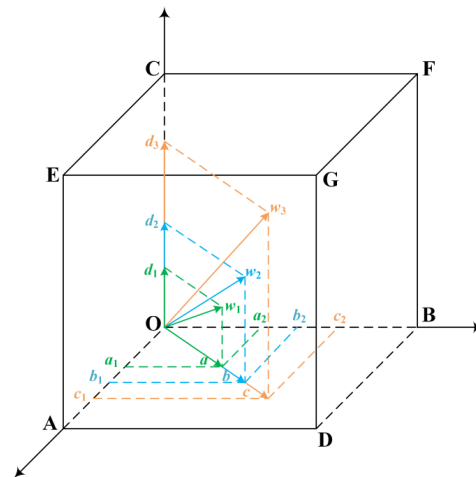


Figure 1. A mechanism diagram of the coordinated development of the EST system.

The operation of EST encompasses not only the three independent subsystems of education, science and technology, and talent, each with unique functions and operational methods distinct from other subsystems, but also the complex relationships between these three subsystems (Figure 2). In the process of the system's operation and development, education, science and technology, and talent are three subsystems that play a crucial role in the stability and security of EST. The normal operation of EST also requires secondary components to complement the overall development of EST, acting as environmental variables for its development. While the science and technology, education, and talent subsystems provide support for other subsystems, they are also influenced by them, forming a mechanism of mutual feedback and development. For instance, the science and technology subsystem continuously supplies new technologies, models, and methods for the development of the education and talent subsystems. Its own development is inseparable from the progress of basic education and the cultivation of high-quality scientific and technological talents. These needs are aligned with the development of the education and science and technology subsystems. Therefore, while each subsystem ensures its independent operation and development, it is also subject to constraints and influences from other subsystems, thus forming a state where both restrictive and promotional effects coexist in the process of system development.

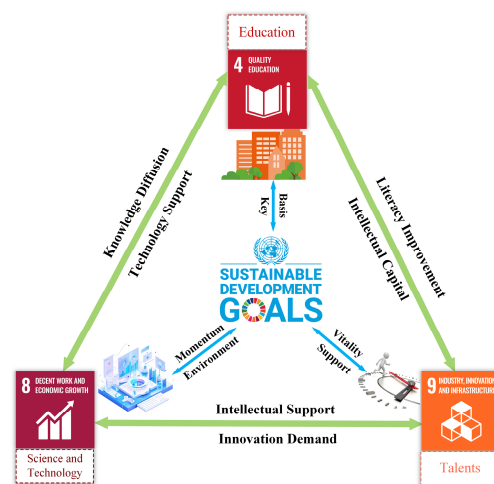


Figure 2. Collaborative development of education, science and technology, and talent.

3. Materials and Methods

3.1. Study Area

In this study, 22 provinces, 4 municipalities directly under the central government (Beijing, Tianjin, Shanghai, and Chongqing), and 4 autonomous regions (Inner Mongolia, Ningxia, Xingjiang, and Guangxi) in China were selected as study areas (Figure 3). Due to data limitations, Tibet, Hong Kong, Macao, and Taiwan were not studied. The research sample in this article covers the time span from 2001 to 2022. In terms of specific regional division, according to the classification standards of the National Bureau of Statistics of China, the overall regional economic pattern of China is divided into three major blocks, namely the eastern regions, the central regions, and the western regions. Due to the macroscopic scale of the research conducted based on the three major blocks, this paper further refines it to the provincial level, conducting research on a relatively smaller scale.



Figure 3. The spatial distribution of the study areas.

3.2. Evaluation Index System

In order to measure the integrated development level of EST across different provinces in China, and further explore the spatiotemporal evolution characteristics of the integrated development level of EST between different regions, this paper constructs a measurement system with 15 indicators from the three dimensions of education, science and technology, and talent to better reflect the differences in the integrated development level of EST among different regions (Table 1).

The selection of measurement indicators for the education subsystem is centered around the main entities such as schools, teachers, and students, and it takes into account the government's investment and attention to education. In addition to indicators that reflect educational funding and the number of higher education institutions, the indicator of educational attention, which is more difficult to directly measure, is also chosen to reflect the emphasis on education during regional development [20]. The science and technology subsystem focuses on selecting measurement indicators from the entire chain and process of technological innovation, considering aspects such as R&D investment, innovation processes, and the transformation and application of innovation outcomes [21]. Moreover, due to the phenomenon in China where innovative activities emphasize the quantity of patent outputs while neglecting quality, the indicator of technological complexity is chosen to measure the progress of innovation quality. The talent subsystem indicators are centered around human beings, on the one hand, depicting the situation of high-quality talents in

various regions, that is, researchers and high-quality talents who undertake intellectual labor [22]. On the other hand, the impact of external factors such as the employment environment and social atmosphere on talent development is considered. The process of knowledge dissemination requires certain media as carriers, and books play an important role in the dissemination of knowledge [23].

Table 1. Evaluation index system of EST level.

1st Index	2nd Index	3rd Index	Calculation Method	Attribute
EST level	Education	Education Investment	Education expenditure/budget expenditure	Positive
		Higher Education	Number of higher education schools/Total number of higher education schools	Positive
		Higher Education Student	Number of higher education admissions/Total population	Positive
		Basic Education	Number of primary school teachers per 10,000 people	Positive
		Education Attention	Reference [24]	Positive
	Science and Technology	R&D Expenditure	Total R&D expenditure/GDP	Positive
		Technology Transaction	Technology market transaction amount/GDP	Positive
		Per Capita Patent Quantity	Total number of patents/Total population	Positive
		Innovation Efficiency	Total factor productivity	Positive
		Technological Complexity	Reference [25]	Positive
	Talent	R&D Personnel Intensity	Number of patent authorizations/Full-time equivalent of R&D personnel	Positive
		Higher Education Faculty	Number of full-time faculty in ordinary higher education institutions/Total population	Positive
		High-Quality Population	Number of higher education students/Total population	Positive
		Per Capita Reading	Public library book collection/Total population	Positive
		Unemployment	Number of unemployed people/Total population	Negative

Notes: The data are from the China Statistical Yearbook (2002–2023), the China Statistical Yearbook of Science and Technology (2002–2023), and the statistical bulletins and statistical data issued by provincial statistical bureaus.

3.3. Component Weights

The variables g_j ($j = 1, 2, 3$) are the parameters of education, science and technology and talents, respectively. It is necessary to standardize the datasets to avoid the influence of measurement methods and units. g_{ij} is the i th index of the j th-order parameter, expressed by X_{ij} ($i = 1, 2, \dots, n$). The calculation process is as follows:

$$g_{ij} = (X_{ij} - \min X_{ij}) / (\max X_{ij} - \min X_{ij}) \text{ Positive} \quad (1)$$

$$g_{ij} = (\max X_{ij} - X_{ij}) / (\max X_{ij} - \min X_{ij}) \text{ Negative} \quad (2)$$

where g_{ij} is the normalized value of each indicator, and its value range is $[0, 1]$. The overall order parameters can be found using the integrated approach. The calculation process is as follows:

$$G_j = \sum_{i=1}^n f_i g_{ij} \quad (3)$$

where G_j is the order degree of education, science and technology, and talent, and λ_{ij} is the parameter weight. f_i is the indicator of comprehensive weight. The AHP and EWM are used to calculate the weights [26]. The least square decision (LSD) model is established to control the deviation of the judgment index decision results in a small range [27]. The calculation equation is as follows:

$$\min H(f) = \sum_{i=1}^m \sum_{j=1}^n \left\{ [(g_j - f_j)X_{ij}]^2 + [(v_j - f_j)X_{ij}]^2 \right\} \quad (4)$$

where subjective weight vector $v = (v_1, v_2, \dots, v_n)^T$, objective weight vector $g = (g_1, g_2, \dots, g_n)^T$, integrated weight vector $f = (f_1, f_2, \dots, f_n)^T$, and $\sum_{i=1}^n f_i = 1, f_i \geq 0 (i = 1, 2, \dots, n)$.

3.4. Coupling Coordination Degree Model

Coupling refers to the extent of mutual influence and interaction among two or more subsystems, and it can also reflect the degree of constraint between different subsystems. The degree of coupling can measure the level of collaborative development between different subsystems and can, to a certain extent, depict the differences in the development levels of different subsystems [28]. This paper employs the degree of coupling to measure the interrelationships between EST. Specifically, the degree of coupling is used to measure the extent of interaction and influence between systems, while the degree of coordination is used to measure the level of collaborative development and evolution. The calculation formula for the degree of coupling is as follows:

$$C = \left[\frac{e_1 \times e_2 \times e_3}{\left(\frac{e_1 + e_2 + e_3}{3} \right)^3} \right]^{\frac{1}{3}} \quad (5)$$

It is important to note that while the coupling degree C can to some extent determine the strength of the interaction between the education, science and technology, and talent subsystems, there may be cases where the subsystems have low development levels but a high coupling degree [29]. This does not truly reflect the overall development level of EST, which is inconsistent with the research theme of this paper. Therefore, on the basis of calculating the coupling degree, this paper introduces a coordination degree model. The calculation method is as follows:

$$D = \sqrt{C \times T} \quad (6)$$

$$T = \alpha e_1 + \beta e_2 + \gamma e_3 \quad (7)$$

e_i represents the development levels of the three subsystems, D represents the coupling coordination degree, and T represents the overall development level of EST. α , β , and γ are constant coefficients whose sum equals 1. The relative importance of each subsystem is considered equal in this study, with education, science and technology, and talent being regarded as three equally important subsystems, each assigned the same weight. In the classification of regional coupling coordination degrees, based on the ArcGIS 10.8 software and using the natural breaks method, this paper divides the coupling coordination degree into three levels, low, medium, and high, to conduct a study on the spatial differentiation characteristics of the EST coupling coordination degree in different regions.

3.5. Exploratory Spatiotemporal Data Analysis

3.5.1. Construction of New Spatial Weighting Matrix

In order to conduct spatiotemporal dynamic transfer analysis, it is necessary to first construct a transition matrix that can reflect spatiotemporal characteristics. Existing research tends to focus on constructing spatial vector weight matrices based on geographical locations, which overlooks the heterogeneity between different regions [30]. Based on this,

this paper further considers the issue of spatial heterogeneity, incorporating economic development factors into the spatial vector weight matrix to construct a composite spatial vector weight matrix [31].

$$\theta_{ij} = \begin{cases} \left(\frac{NL_i}{NL_j}\right)^{\frac{1}{2}} \times \frac{1}{d_{ij}} & (i \neq j) \\ 0 & (i = j) \end{cases} \quad (8)$$

where θ_{ij} is the spatial weight of spatial units i and j . d_{ij} is the distance between spatial units i and j . NL represents the economic development level.

3.5.2. LISA Time Path

Based on the ESTDA model, the LISA time path incorporates the temporal dimension into the LISA to realize dynamic interaction [32]. The Markov transfer matrix is used to reveal the spatiotemporal interaction under geographical elements. The LISA time path can be calculated by relative length (Equation (9)) and tortuosity (Equation (10)):

$$RL_i = \frac{n \times \sum_{t=1}^{T-1} d(L_{i,t}, L_{i,t+1})}{\sum_{i=1}^n \sum_{t=1}^{T-1} d(L_{i,t}, L_{i,t+1})} \quad (9)$$

$$D_i = \frac{\sum_{t=1}^{T-1} d(L_{i,t}, L_{i,t+1})}{d(L_{i,t}, L_{i,T})} \quad (10)$$

where n is the number of areas; T is the time period; $L_{i,t}$ represents the area i in Moran's I scatter diagram in t year; $d(L_{i,t}, L_{i,t+1})$ represents the distance that region i moves from year t to year $t + 1$.

3.5.3. LISA Spatiotemporal Transition

LISA is used to express the spatial features of geographical elements. Rey et al. divided the transition into $Type_0$, $Type_1$, $Type_2$, and $Type_3$ [33]. Equations (11)–(13) are as follows:

$$SF = \frac{Type_1 + Type_2}{m} \quad (11)$$

$$SC = \frac{Type_0 + Type_{3A}}{m} \quad (12)$$

$$p = 1 - \frac{\sum p_{i,i}}{k} \quad (13)$$

where SF and SC represent the spatiotemporal flow and condensation of EST. $Type_0$, $Type_1$, $Type_2$, and $Type_{3A}$ represent the transition numbers; p represents the relative movement rate. $p_{i,i}$ represents the diagonal elements of the spatiotemporal transition matrix, $k = 4$. When $p = 1$, all regions have state transitions; when $p = 0$, no region has a transition (Table 2).

Table 2. Types of LISA spatiotemporal transitions.

Type	Type of Change	Specific Meaning
$Type_0$	$LH_t \rightarrow LH_{t+1}, HL_t \rightarrow HL_{t+1}, HH_t \rightarrow HH_{t+1}, LL_t \rightarrow LL_{t+1}$	Local and adjacent do not transfer
$Type_1$	$LH_t \rightarrow HH_{t+1}, HL_t \rightarrow LL_{t+1}, HH_t \rightarrow LH_{t+1}, LL_t \rightarrow HL_{t+1}$	Local transfer and adjacent do not transfer
$Type_2$	$LH_t \rightarrow LL_{t+1}, HL_t \rightarrow HH_{t+1}, HH_t \rightarrow HL_{t+1}, LL_t \rightarrow LH_{t+1}$	Local do not transfer and adjacent transfer
$Type_3$	$HL_t \rightarrow LH_{t+1}, LH_t \rightarrow HL_{t+1}, LL_t \rightarrow HH_{t+1}, LL_t \rightarrow LL_{t+1}$	Local and adjacent transfer

Notes: If the transfer direction of the local and adjacent is the same, it is called $Type_{3A}$; otherwise, it is called $Type_{3B}$.

4. Results and Discussion

4.1. The Temporal Characteristics of the Integrated Development Level of EST in China

To further analyze the temporal characteristics of education, science and technology, and talent in different regions, Figure 4 illustrates the trend in the development levels of education, science and technology, and talent in the eastern, central, and western regions, as well as nationwide. From a national perspective, during the research period, the development levels of education, science and technology, and talent have generally shown an upward trend. The development levels of the education, science and technology, and talent subsystems show a decreasing order, indicating that China's educational development level is greater than that of science and technology and talent. Looking at the regional development of education, science and technology, and talent, the eastern region's development levels of education, science and technology, and talent have shown a stable upward trend during the research period, rising from 0.237, 0.276, and 0.217 in 2001 to 0.539, 0.448, and 0.380 in 2022, with increases of 127.17%, 61.96%, and 75.71%, respectively. The important role of education in scientific and technological innovation and talent cultivation is not limited to China; this perspective also applies to Europe. In November 2020, the European Commission published the "Recommendation on vocational education and training (VET) for sustainable competitiveness, social fairness and resilience" [34]. This recommendation posits that the future development of scientific and technological innovation requires education and training as a crucial foundation, especially in the era of the digital economy. The developmental threshold effect brought about by the growth of the digital economy compels various groups to engage in lifelong learning to meet the developmental demands of the digital economy era and the needs of enterprise digital transformation.

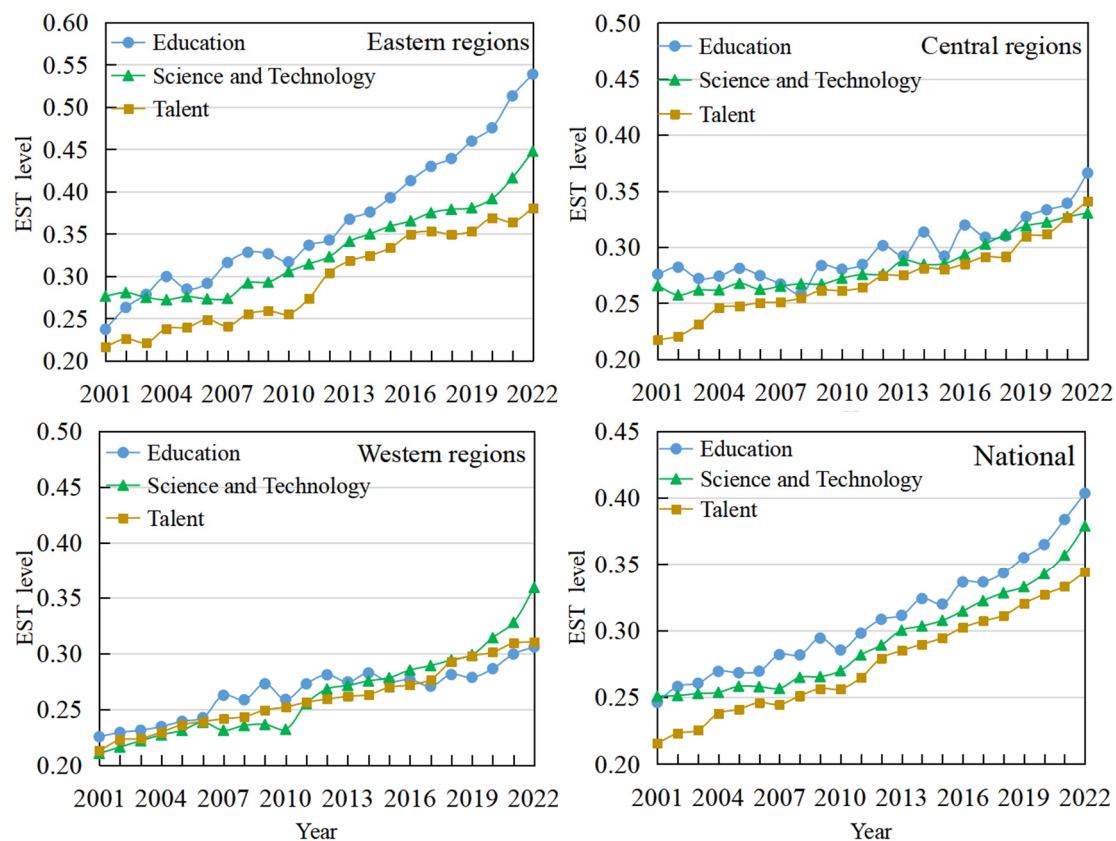


Figure 4. Change trend in education, science and technology, and talent from 2001 to 2022.

In terms of growth rate, the education subsystem has the fastest development speed, while the science and technology subsystem has the slowest. Compared with the eastern region, the development levels of the education, science and technology, and talent sys-

tems in the central and western regions have certain fluctuations, but overall, they have maintained an upward trend. In terms of growth rate, the talent subsystem in the central region has the fastest development speed, rising from 0.217 in 2001 to 0.341 in 2022, with an increase of 56.96%. The science and technology subsystem in the western region has the fastest development speed, rising from 0.210 in 2001 to 0.359 in 2022, with an increase of 70.61%. During the research period, the aforementioned development trends were observed. The specific reasons for this may be as follows: Firstly, the Chinese government has continuously increased its investment in education, science and technology, and talent development, promoting a continuous rise in the development levels of these areas. On the other hand, different regions in China have varying foundations for education, science and technology, and talent development, resulting in distinct developmental levels among the eastern, central, and western regions during the development process.

4.2. The Spatial Characteristic of the Level of EST in China

This section, based on ArcGIS 10.8 software and using the natural breaks method, mapped the spatial distribution of the coupling coordination degree of EST. Figure 5a–d represent the spatial distribution maps of the coupling coordination degree of EST for the years 2001, 2008, 2015, and 2022, respectively.

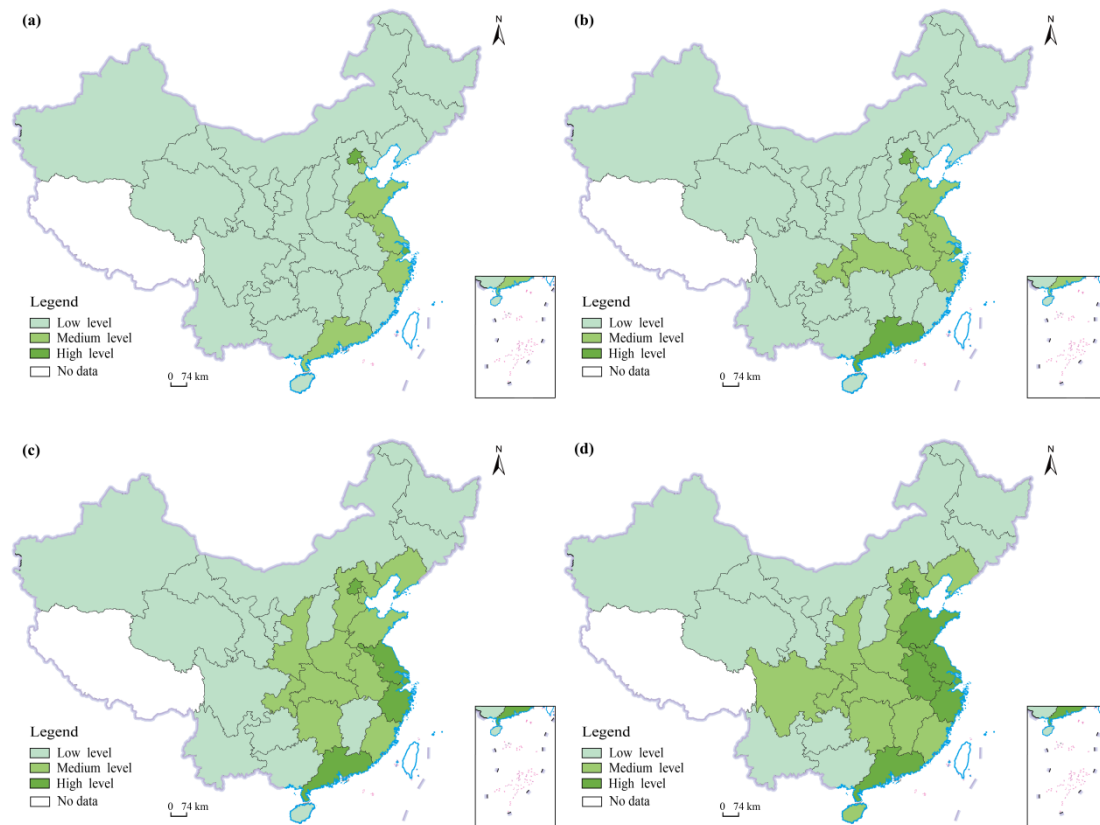


Figure 5. The spatial distribution of the EST level in (a) 2001, (b) 2008, (c) 2015, and (d) 2022.

According to Figure 5, the number of areas with medium and high levels of coupling coordination degree has been continuously increasing, gradually spreading from coastal regions to inland areas during the research period. Beijing and Shanghai, as the political and economic centers of China, have consistently maintained a high level of coupling coordination development throughout the research period. The number of areas with a high level increased from 2 in 2001 to 7 in 2022, and those with a medium level increased from 5 in 2001 to 11 in 2022, while the number of areas with a low level decreased from 24 in 2001 to 10. Tianjin, Shandong, Jiangsu, Zhejiang, and Guangdong, as five major economic

provinces in China, possess superior resources in education, science and technology, and talent. At the same time, China has supportive policies for the advantageous development of coastal regions. Tianjin, Shandong, Jiangsu, Zhejiang, and Guangdong have leveraged their multiple advantages for rapid development and have consistently maintained a leading edge.

This indicates that the coupling coordination development of EST in different regions of China exhibits significant spatial differentiation characteristics across different years. The change in the level of coupling coordination development shows a trend from coastal to inland areas, which is basically consistent with the evolution of China's economic development pattern. With the rapid development of China's central and western regions, the eastern regions, while accumulating a large amount of educational, scientific and technological, and talent resources, have also fully exerted their spillover effects, leading to an increase in the EST levels in the central and western regions. This has resulted in a significant number of areas in central and western China upgrading their EST levels from low to medium categories between 2001 and 2022.

4.3. The Spatiotemporal Interaction Characteristics of China's EST

4.3.1. Geometric Characteristics of LISA Time Path

According to Equations (9) and (10), this paper calculates the relative length and curvature of the LISA (Local Indicators of Spatial Association) temporal path for the coupling coordination degree of EST. The natural breaks method is used to categorize them into three classes (high relative length, medium relative length, and low relative length; high curvature, medium curvature, and low curvature). The division of transition directions mainly employs the quadrant method, the directions of which are at 90° , 180° , 270° , and 360° . Here, 0° to 90° indicates a collaborative positive growth trend between the local area and its neighbors, 90° to 180° and 270° to 360° suggest a negative growth trend between the local area and its neighbors, and 180° to 270° represents a collaborative negative growth trend.

As can be seen from Figure 6a, the number of provinces where the relative length of the LISA temporal path for the coupling coordination degree of EST is less than the average exceeds 50%, indicating that the spatial pattern of EST development has strong stability. The high-value areas of the relative length of the EST temporal path are mainly concentrated in the northeast, northwest, and southwest regions, while the low-value areas are concentrated in the eastern coastal areas. This is mainly because the eastern coastal areas have a higher level of economic development, with ample financial support for the development of the education, science and technology, and talent systems, and a good base for the development of education, science and technology, and talent, which makes the spatial pattern of EST more stable.

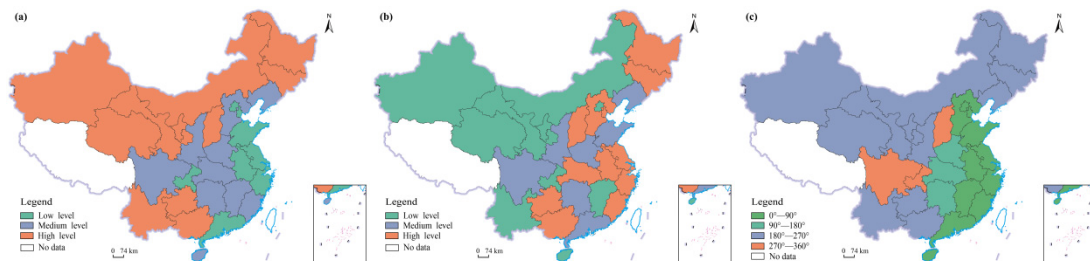


Figure 6. Spatial distribution of length (a), tortuosity (b), and movement direction (c) of LISA time path of EST.

According to Figure 6b, overall, the curvature of the LISA temporal path for the coupling coordination degree of EST shows a decreasing trend from the coastal areas to the inland and from the southeast to the northwest. During the research period, the curvature of the LISA temporal path for the coupling coordination degree of EST was greater than

1 in 17 provinces, accounting for more than 50% of the total research units, indicating that the LISA temporal path curvature of EST in most provinces in China showed a certain degree of change during the research period, suggesting that different provinces were influenced by neighboring areas to some extent. During the research period, the LISA temporal path curvature was at a low level in 11 provinces (Xinjiang, Inner Mongolia, Gansu, Ningxia, Qinghai, Beijing, Shanghai, Chongqing, Yunnan, Jiangxi, Hainan) and at a medium level in 6 provinces (Liaoning, Shandong, Shaanxi, Sichuan, Hunan, Guangdong), indicating that most provinces have relatively stable characteristics in the local spatial growth process and dependency direction.

According to Figure 6c, during the research period, 12 provinces (Beijing, Tianjin, Hebei, Shandong, Jiangsu, Anhui, Zhejiang, Fujian, Jiangxi, Guangdong, Hainan, Shanghai) exhibited a collaborative positive growth trend, primarily distributed in the eastern region of China. Conversely, 12 provinces (Heilongjiang, Jilin, Liaoning, Inner Mongolia, Xinjiang, Gansu, Ningxia, Qinghai, Shaanxi, Yunnan, Guizhou, Guangxi) showed a collaborative negative growth trend, mainly concentrated in the northwest, northeast, and southwest regions. Additionally, six provinces (Henan, Hubei, Hunan, Shaanxi, Sichuan, Chongqing) exhibited a non-collaborative growth trend, primarily located in the central region. Overall, the spatial pattern of EST is relatively stable. The spillover effects in provinces such as Beijing, Shanghai, Shandong, Jiangsu, Zhejiang, and Guangdong in eastern China are quite pronounced, driving the development of EST in surrounding provinces and demonstrating a clear collaborative high-growth trend. In contrast, the northeastern, northwestern, and southwestern regions of China, lacking effective stimulation from core cities and rapidly developing areas, have a generally low level of development for EST, exhibiting a distinct collaborative low-growth trend.

4.3.2. Spatiotemporal Transition Characteristics of LISA

The LISA spatiotemporal transition can better describe the spatial association and dynamic transition characteristics between different geographical units [35]. Therefore, this study calculates a spatiotemporal transition probability matrix to further study the transition characteristics of the local spatial correlation of China's EST.

According to Table 3, it can be seen that the most frequent transition type for EST and its subsystems of education, science and technology, and talent is $Type_0$, with respective proportions of 0.802, 0.767, 0.814, and 0.743. The probability of Moran's I scatterplot being in the same quadrant is over 70%, indicating that most provinces did not experience significant status transitions during the research period. The spatiotemporal clustering for EST and its subsystems is over 80%, and the spatiotemporal mobility is less than 0.2, suggesting a pronounced spatiotemporal inertia in EST and its subsystems. This implies that the spatial distribution patterns of the elements have path dependence and locking characteristics. Regarding the relative mobility probability, the relative mobility probabilities (p) for EST and its subsystems are all less than 0.1, further indicating that EST and its subsystems have strong spatial dependencies, and their spatial patterns are relatively stable.

Table 3. Spatiotemporal transition matrix of education, science and technology, talent, and EST.

	$t/t + 1$	HH	LH	LL	HL	Type	n	Proportion	SF	SC	p
EST	HH	0.895	0.078	0.004	0.023	$Type_0$	554	0.802	0.160	0.849	0.089
	LH	0.009	0.912	0.016	0.063	$Type_1$	68	0.098			
	LL	0.032	0.011	0.922	0.035	$Type_2$	42	0.062			
	HL	0.025	0.055	0.004	0.916	$Type_3$	26	0.038			
Education	HH	0.952	0.004	0.040	0.004	$Type_0$	529	0.767	0.149	0.818	0.059
	LH	0.006	0.928	0.066	0.000	$Type_1$	56	0.081			
	LL	0.000	0.000	0.917	0.083	$Type_2$	47	0.068			
	HL	0.032	0.002	0.002	0.964	$Type_3$	58	0.084			

Table 3. Cont.

	$t/t + 1$	HH	LH	LL	HL	Type	n	Proportion	SF	SC	p
Science and technology	HH	0.856	0.023	0.104	0.017	Type ₀	561	0.814	0.108	0.844	0.074
	LH	0.004	0.963	0.033	0.000	Type ₁	43	0.062			
	LL	0.064	0.005	0.931	0.000	Type ₂	32	0.046			
	HL	0.000	0.005	0.041	0.955	Type ₃	54	0.078			
Talent	HH	0.955	0.025	0.020	0.000	Type ₀	513	0.743	0.198	0.811	0.068
	LH	0.000	0.962	0.000	0.038	Type ₁	49	0.071			
	LL	0.132	0.039	0.829	0.000	Type ₂	87	0.127			
	HL	0.016	0.000	0.000	0.984	Type ₃	41	0.059			

4.4. Discussion of Results

The main focus of this section is to compare the findings of this paper with those of existing research articles, with the primary aim of accurately positioning this paper within the literature. Based on this consideration, this section will concentrate on comparing this paper with the existing research literature in terms of research content, methods, and conclusions. The goal is to identify similarities with other papers that are closely related to this one, to understand the reasons for any differences.

In terms of research content, the comprehensive consideration of the three subsystems of education, science and technology, and talent is a significant similarity between the existing literature and the research presented in this paper. Some studies begin by analyzing the coupling relationships among the three subsystems of education, science and technology, and talent, proposing that the integrated development of education, science and technology, and talent has theoretical, historical, and practical logics [36]. In the process of promoting the coupling of education, science and technology, and talent, the three elements exhibit a “chain” relationship, maintaining the operation and development of the system as a whole. This is similar to the theoretical analytical framework established in this paper. However, in terms of the methodology for expanding the analysis of the main research content, some studies conducted qualitative research, which involves logical reasoning based on the existing literature and reality [36,37]. This differs from the approach taken in this paper, which combines qualitative and quantitative analysis.

In terms of research methodology, existing studies employed a variety of approaches, primarily categorized into qualitative and quantitative research methods. As previously mentioned, some studies utilized quantitative research, while others also employed qualitative methods [38]. The research on complex systems shares some similarities with the methods used in this paper, such as the construction of an evaluation index system, the selection of measurement models, and the collection and analysis of relevant data [11,24]. The difference lies in the fact that this paper introduces a composite spatial vector weight matrix that considers both geographical and economic dimensions and conducts a feature analysis of spatiotemporal transitions. In contrast, existing studies primarily focused on the static level of development, that is, the coupling and coordination of the development of the three subsystems of education, science and technology, and talent in a given year.

Due to the variety of research methods employed in existing studies, there is a focus on different aspects during the analysis. Consequently, the conclusions drawn are also varied. From the qualitative research findings, there is relative consistency with the research presented in this paper. Different research outcomes have indicated that there is a very complex coupling and coordination relationship among the three subsystems of education, science and technology, and talent [36,37]. This is primarily due to the interplay, influence, and constraints among different subsystems. The development of one subsystem is affected by the development levels of the others, and no subsystem exists in isolation within the overall development context.

However, there are certain differences between the conclusions drawn from qualitative research and the findings of this paper. For instance, this study finds that the educational subsystem is in the leading position within the entire system, while another empirical

study suggests that talent is in the dominant position [38]. The differences observed may be attributed to several factors: On one hand, the time periods selected for the two studies are different, and there are significant differences in the development levels of the education, science and technology, and talent subsystems between different years. On the other hand, the focus of the selected indicators varies, which may lead to divergent conclusions.

5. Conclusions and Practical Implications

5.1. Conclusions

This study investigates the development level and spatiotemporal pattern evolution of China's EST from 2001 to 2022 by constructing an index system. This paper provides a comprehensive and systematic theoretical framework for the study of China's EST. The conclusions are as follows:

(1) From 2001 to 2022, the development levels of education, science and technology, talent, and EST in China have generally shown an upward trend. Looking at the evolutionary perspective of the time series, different provinces have varied growth rates, but overall, they maintain an upward momentum.

(2) The spatial differentiation characteristics of China's EST are very pronounced. The rise in EST levels is mainly driven by the education subsystem, with the science and technology and talent subsystems playing a secondary role in driving EST levels. There are regional disparities in EST levels among different provinces, and these regional disparities are also changing over time.

(3) The analyses of the geometric characteristics of the LISA time path and the spatiotemporal transition show that China's EST generally presents a relatively stable spatial characteristic. During the research period, although the development of EST experienced certain spatiotemporal fluctuations, it was generally stable overall.

5.2. Practical Implications

On the basis of revealing the development level and spatiotemporal evolution pattern of China's EST as discussed earlier, this paper further proposes policy implications for the shortcomings and weak links in the development of China's EST. This study proposes the following practical implications:

(1) At the national or regional level, it is imperative to promptly introduce medium- and long-term planning schemes that serve the development of the three subsystems: education, science and technology, and talent. These plans should be designed from the perspective of long-term government planning to guide the coordinated development of the education, science and technology, and talent subsystems. Although the development level of China's EST shows an overall upward trend, the driving forces of each subsystem vary. Therefore, the development of China's EST should continue to strengthen the foundation of basic education, higher education, and quality education and further enhance the input in the science and technology subsystem and the talent subsystem. On this basis, it is necessary to further enhance the driving capacity of the science and technology subsystem and the talent subsystem for the development of EST.

(2) In the future design of government regional coordination development policies, EST should be considered as a key lever to achieve coordinated regional development. Given the significant spatial differentiation characteristics of China's EST development, it is essential to focus on regional coordinated development policies to reduce the disparities in EST development levels between regions. The government should increase the intensity of transfer payments to ensure the funding needed for the development of education, science and technology, and talent in underdeveloped areas. On this basis, underdeveloped regions are encouraged to fully leverage the spillover and driving effects of developed provinces, to achieve the sharing of EST development experiences and models, and to narrow the development gap in EST among different regions.

(3) The development of China's EST exhibits a significant degree of path dependence across various provinces. While this maintains stability in the developmental landscape, it

is not conducive to exploring new developmental pathways. Therefore, it is essential to build upon the existing developmental pathways by further enhancing the regions' original innovation capabilities and strengthening the driving force of basic research in talent development and innovation. It is also essential to continuously seek new developmental pathways beyond the current scope of research.

5.3. Limitations and Future Research

This paper also has some shortcomings and directions for future expansion. First, the research object of this paper consists of the 30 provinces in mainland China. Future research can further refine the research object to more microscopic scales such as prefecture-level cities, rural areas, and villages. It is also possible to build a cross-country panel dataset based on the research of this paper and conduct empirical studies on different countries. Second, this paper focuses more on the evaluation of the development level of the three subsystems of education, science and technology, and talent, with relatively less discussion on the driving mechanism. Future research can focus on the influencing factors of the development level of EST and diagnose the obstacles to EST development. Third, the research paradigm used in this paper is mainly the classic economic geography paradigm. Against the background of the rapid development of new-generation information technologies such as big data and cloud computing, research methods such as machine learning can be used to improve and develop the existing paradigm.

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